BT603 – (Fermentation Technology)

Current + Past Papers Solved By ~Bella♡

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♡ Subjective ♡

Types of fermentation on the basis of Oxygen? 5 Marks

The O2 supply must be according to need of the process of fermentation:

- **Aerobic Fermentation**

Aerobic fermentation is carried out in the presence of oxygen organisms use oxygen for the conversion of complex organic compounds. Oxygen here is actually required for the reproduction and growth of microorganisms.

- **Semi-anaerobic Fermentation**

It is a process in which organism makes ATP by aerobic respiration, but capable of switching to anaerobic respiration if oxygen is absent. Oxygen is needed, but not much more.

- **Anaerobic Fermentation**

Fermentation is carried out in the strict absence of oxygen means presence of oxygen may be toxic. Involves conversion of complex organic compounds into simpler ones in the absence of oxygen.

Brief note on moleases 5 marks

Molasses is used in the production of high-volume/low-value products such as ethanol, SCP, organic and amino acids and some microbial gums. In 1980, 300,000 tons of cane molasses were used for amino acid production in Japan (Minoda, 1986). Molasses or sucrose also may be used for production of higher value/low-bulk products such as antibiotics, specialty enzymes, vaccines and fine chemicals. The cost of molasses will be very competitive when compared with pure carbohydrates. However, molasses contains many impurities and molasses-based fermentations will often need a more expensive and complicated extraction/purification stage to remove the impurities and effluent treatment will be more expensive because of the unutilized waste materials which are still present.

What is the definition of fermentation?

Fermentation is a metabolic process that releases energy from sugar or other organic substrates through various enzyme actions. It is a type of anaerobic biochemical process.

Procedure of fermentation? 5 mark

Fermentation is an anaerobic biochemical process. In fermentation, the first process is the same as cellular respiration, which is the formation of pyruvic acid by glycolysis where net 2 ATP molecules are synthesized. In the next step, pyruvate is reduced to lactic acid, ethanol or other products. Here NAD+ is formed which is re-utilized back in the glycolysis process.

What the types of fermentation on the basis of the products?

Following are the types of fermentation on the basis of the products that formed:

1. Beer Fermentation
2. Yogurt Fermentation

3. Citric acid Fermentation
4. Pickle Fermentation and many more.

Difference between batch culture and continuous culture?

Batch Culture

- It is a closed batch.
- All the nutrients are added at once in the beginning.
- The product is extracted at the end of the batch.
- The productivity level is low.
- It is used to extract secondary metabolites such as antibiotics.
- The growth conditions can be changed by altering pH and dissolved oxygen.
- Large fermenters are used.
- It has less chance of contamination.
- The microbes go through three phases – lag, exponential and stationary.
- Nutrients are consumed at a slower rate.

Continuous Culture

- It is an open batch.
- Nutrients are added throughout the batch.
- The product is continuously removed while the batch is operating.
- The productivity level is high.
- It is used to extract primary metabolites such as amino acids and proteins.
- The growth conditions have to be maintained throughout the operation of the batch.
- Small-size fermenters are used.
- It has a high chance of contamination.

- The microbes go through a very short lag phase and then maintain themselves at the exponential phase.
- Nutrients are consumed rapidly.

Basic components of fermentation?

1. Organism (Hero of Fermentation process)
2. Fermenter (stage on which microorganism act/grow).
3. Media (Formulation)
4. Sterilization.
5. Monitoring/Control (Environmental conditions).
6. Recovery of the product (Isolation, purification etc).
7. Effluent treatment (processing of materials that remained after fermentation process.

What are the micro and macro minerals used in fermentation process?

In many media, magnesium, phosphorus, potassium, Sulphur, calcium and chlorine are components, and because of the concentrations required, they must be added as components. Others such as cobalt, copper, iron, manganese, molybdenum and zinc also essential but are usually present as impurities in other major ingredients.

Difference between rude media and defined media? 3 Marks

Crude media

- ✓ Crude media is made up of unrefined agricultural products e.g. containing barley.
- ✓ Crude media is cheap but composition is variable.
- ✓ Crude media is used for large volume inexpensive products e.g. biofuel from whey

Defined media

- ✓ Defined media are like those we use in the lab e.g. minimal salts medium.
- ✓ Defined media is expensive but composition is known and should not vary.
- ✓ Defined media is used for expensive low volume products e.g. anticancer drugs.

Types of fermentation on the basis of scale? 3 Marks

On the basis of scale there are following types of fermentation:

- ✓ Laboratory scale Fermentation
- ✓ Pilot scale Fermentation
- ✓ Industrial scale Fermentation

Laboratory scale fermentation is carried out in flasks, and cultures may be mixed by means of shaking in a controlled environment chamber.

Pilot and industrial scale fermentations are normally carried out in stirred, aerated vessels, termed as fermenters.

Write the two modes of sterilization in fermentation?

Mode of sterilization

In-situ -> means sterilization is done on the spot/inside fermenter maintaining high temperature by steam.

Ex-situ -> means sterilization is done outside e.g. in autoclave.

Difference between lab scale and pilot scale fermentations?

- **Lab Scale Fermentation**

This is small scale fermentation mostly run at lab scale either in flask of different quantity or fermenter.

- ✓ Sterilization mode at this scale is always ex-situ.
- ✓ Inoculum that runs fermentation process is prepared in slants/small flasks.

- **Pilot Scale Fermentation**

A pilot scale plant offers flexibility versatility and a broad range of equipment to rapidly convert laboratory results into industrial processes. In fact pilot plant fermentation 'mimic' the industrial scale fermentation process by carrying out the industrial process on a smaller scale.

What is the meaning of Scale-up?

Scale-up means increasing the production scale. For example from the laboratory scale to the pilot plant scale or from the pilot plant scale to industrial scale.

Requirements of microorganism growth?

All micro-organisms require water, Sources of energy, Carbon, Nitrogen, Mineral elements, and possibly vitamins, plus oxygen of aerobic) such as CO2 etc.

Selective media and its examples?

Selective Media

Selective media are designed to suppress the growth of unwanted bacteria and encourage the growth of the desired microbes.

Examples of selective media:

- Bismuth sulfite is one medium used to isolate the typhoid bacterium, the gram-negative Salmonella typhi. Bismuth sulfite inhibits gram-positive bacteria and most gram-negative intestinal bacteria (other than S. typhi), as well.
- Sabouraud's dextrose, which has a pH of 5.6, is used to isolate fungi that outgrow most bacteria at this pH.
- Eosin Methylene Blue Agar (EMB)
- Hektoen Enteric (HE) Agar
- MacConkey Agar

What is role of ph in fermentation?

pH is an important factor in fermentation because it can affect the rate and efficiency of the fermentation process as well as the quality of the final product. However, the optimal pH for a particular fermentation process will depend on the specific microorganisms that are being used and the desired end product. For example, some microorganisms may require a higher pH in order to grow and produce the desired product, while others may prefer a lower ph.

Write down basic fermentation step in large scale?

1. Inoculum Preservation
2. Inoculum Build-up
3. Pre-Fermenter Culture
4. Production Fermentation

Estimation of carbon formula calculation? 2 Marks

$CL = -1/kL^{a} \{(dCL/dt) + xQO^{2})\} + C^{*}$

Limitation of starch?

- Starch has to be broken down into sugar. The sugar then has to be broken down into simple sugars to allow the yeast to react with these sugars during the fermentation (rising).
- We can't ferment starch naturally because starches are too complex for the yeast to digest. They are made up of multiple simple sugars combined.

What are bio fertilizers? 3

Bio fertilizers are low cost, renewable sources of plant nutrients which supplement chemical fertilizers. These are nothing but selected strains of beneficial soil microorganisms cultured in the laboratory and packed in a suitable carrier. They can be used either for seed treatment or soil application.

Fed-Batch Fermentation? 2 Marks

An enhancement of the closed batch process is the fed-batch fermentation. In fed-batch to fermentations, sterile culture medium is added either continuously or periodically to the inoculated fermentation batch.

What are bioweapons? Also write three examples.5

Bioweapons

Biological weapons (often termed “bio-weapons”, “biological threat agents”, or “bio-agents”) are living organisms or replicating entities (viruses, which are not universally considered “alive”) that reproduce or replicate within their host victims. The North Atlantic Treaty Organization (NATO) defines a biological weapons as “the provision of any infectious agent or toxin by any means of delivery in order to cause harm to humans, animals, or plants.”

Examples of Bioweapons

- Smallpox
- Anthrax
- Botulinum toxin
- Aflatoxin
- Clostridium Perfringens

Advantages/Importance of Oils and fats in fermentation?

Oils were first used as carriers for antifoams in antibiotic processes. Vegetable oils (olive, maize, cotton seed, linseed soya bean, etc.) may also be used as carbon substrates, particularly for their content of the fatty acids, oleic, linoleic and linoleic acid, because costs are competitive with those of carbohydrates. Bader et al, (1984) discussed factors favoring the use of oils instead of carbohydrates. A typical oil contains

approximately 2.4 times the energy of glucose on a per weight basis. Oils also have a volume advantage as it would take 1.24 dm^3 of soya bean oil to add 10 kcal of energy to a fermenter, whereas it would take 5 dm^3 of glucose or sucrose assuming that they are being added as 50% w/w solutions. Ideally, in any fermentation process, the maximum working capacity of a vessel should be used. Oil based fed-batch fermentations permit this procedure to operate more successfully than those using carbohydrate feeds where a larger spare capacity must be catered for to allow for responses to a sudden reduction in the residual nutrient level Oils also have antifoam properties which may make downstream processing simpler, but normally they are not used solely for this purpose.

What is the importance of enrichment culture in fermentation? 3 Mark's

Enrichment culture is used when bacteria present in small numbers can be missed, especially if other bacteria are present in much larger numbers, it is sometimes necessary to use an enrichment culture. This is often the case for soil samples. The medium (enrichment medium) for an enrichment culture is usually liquid and provides nutrients and environmental conditions that favor the growth of a particular microbe but not others. In this sense, it is also a selective medium, but it is designed to increase very small numbers of the desired type of organism to detectable levels.

What are the micro organisms in fermentation?

Bacillus, Lactobacillus, Micrococcus, Streptococcus, Pediococcus, and Leuconostoc are the common fermenting bacteria. Moreover, Cladosporium, Aspergillus, Trichothecium, Paecilomyces, Fusarium, and Penicillium are the common fungal genera found in fermented products.

What are the factors that influencing the choice of carbon source?

The rate at which carbon source is metabolized influence the formation of biomass or production of primary or secondary metabolites. Fast growth due to high concentrations of rapidly metabolized sugars is often associated with low productivity of secondary metabolites. The main product of a fermentation process will often determine the choice of carbon source, particularly if the product results from the direct dissimilation of it. In fermentations such as ethanol or single-cell protein production where raw materials are 60 to 77% of the production cost, the selling price of the product will be determined largely by the cost of the carbon source media for Industrial Fermentations. The method of media preparation, particularly sterilization, may affect the suitability of carbohydrates for individual fermentation processes. It is often best to sterilize sugars separately because they may react with ammonium ions and amino acids to form black nitrogen containing compounds which will partially inhibit the growth of many micro-organisms. Starch suffers from the handicap that when heated in the sterilization process it gelatinizes, giving rise to very viscous liquids, so that only concentrations of up to 2% can be used without modification. The choice of substrate may also be influenced by government legislation. The purity of the carbon source may also affect the choice of substrate. For example, metallic ions must be removed from carbohydrate sources used in some citric acid processes.

Carbohydrates composition for Barley Malt? 5 marks

Barley grains may be partially germinated and heat treated to give the material known as malt, which contains a variety of sugars besides starch. Malt is the main substrate for brewing beer. Malt extracts may also be prepared from malted grain.

Write the 5 name of recombinant products? 5 Mark's

- Interferons

- Insulin
- Human Serum Albumin
- Clotting Factors VIII and IX
- Epidermal Growth Factor
- Calf Chymosin
- Bovine Somatostatin

What are common fermentation products? 2 Marks

Acids, Enzymes, Gasses, Alcohols, Hormones, Single Cell Protein, Yogurt, Bread & Pickles, etc.

Advantages of Undefined media? 5 Marks

As a consequence these variations in composition small yield improvements are difficult to detect. Undefined media often make product recovery and effluent treatment more problematical because nor all the components of a complex nutrient source will be consumed by the organism. The residual components may interfere with recovery and contribute to the BOD (Biological Oxygen demand) of the effluent.

Significance of hydrocarbon and their derivatives in industrial fermentation? 2 Marks

It was considered that hydrocarbons and their derivatives might have a potential role as feedstock's in the microbial production of higher value products such as intermediates, pharmaceuticals, fine chemicals and agricultural chemicals.

Uses of water in industrial fermentation? 5 Marks

Water is the major component of all fermentation media, and is needed in many of the ancillary services such as

- Heating

- Cooling
- Cleaning
- Rinsing

When assessing the suitability of a water supply it is important to consider pH, dissolved salts and effluent contamination. The mineral content of the water is very.

In-situ sterilization? 2 Marks

In-situ means sterilization is done on the spot/ inside fermenter maintaining high temperature by steam.

Name two important precursors in fermentation? 2 Marks

- Phenoxy acetic acid
- Cyanides

Six trace elements?

The role of trace elements in medium formulation can be significant. Cultured cells normally require Fe, Zn, Cu, Se, Mn, Mo and B. These are often present as impurities in other media components.

Types of fermentation on the basis of water? 5 Marks

On the basis of water/ moisture requirement process can be:

- **Solid-State Fermentation (SSF)**

In solid-state fermentation, the microorganisms grow on a moist solid with little or no 'free' water, although capillary water may be present. Examples of this type of fermentation are seen in mushroom cultivation, bread-making and the processing of cocca and in the manufacture of some traditional foods, e.g. miso (soy paste), soy sauce, tempeh (soybean cake) and gari (cassava), which are now produced in large industrial operations. SSF is also called as Koji Fermentation.

- **Submerged/Liquid Fermentation (SmF)**

Fermentation process wherein microorganisms employed grows in

Submerged state within the fermentation media. Submerged fermentation may use a dissolved substrate e.g. sugar solution, or a solid substrate suspended in a large amount of water to form a slurry. Organism is grown in a liquid medium which is vigorously aerated & agitated. Used for pickling vegetables, brewing beer and producing wine and soy sauce.

Surface Culture/ Jelly Fermentation

It refers to those fermentations in which the microorganisms utilized to grow on the surface of the fermentation media. Organism is allowed to grow on the surface of a liquid medium without agitation. After an appropriate incubation period the culture filtrate is separated from the cell mass and is processed to recover the desirable product. Citric acid is commonly produced by surface fermentation process wherein the microorganism Aspergillus Niger grows as a thick floating mycelial mat over the surface of the media used.

How many types of fermentations are classified?

Fermentation can be classified into different types on the basis of followings:

1. Oxygen (O2) Need
2. Moisture/Water (H20) content in medium
3. Production Scale
4. Process
5. State/Mixing
6. Product
7. Organism

Types of fermentation on the basis of mixing?

On the basis of state or mixing fermentation can be

- **Static Fermentation**

In this type of fermentation there is no mixing during the process. Surface culture un-aerated and unmixed bioreactor can be used here. Un-aerated and unmixed bioreactor is the one in which the bed is static, or mixed only very occasionally (i.e., once or twice per day), and air is circulated around the bed but not blown forcefully.

- **Shake-Flask Fermentation**

On small level, simple shaking procedure can be applied during fermentation process, hence termed as shake flask fermentation. Shake culture flasks are usually applied to aerobic processes mentation. Shake cultures are a convenient method of growing microorganisms in submerged cultures under aerobic conditions created by shaking; it is a small scale equivalent of stirred tank bioreactor.

- **Stirred Fermentation**

This type of fermentation involves proper mixing during the process of fermentation which is done by agitator/ Impeller. The medium must be stirred to keep the cells in suspension various types of stirrers range from simple magnetic stirrers, flat blade turbine impellers, to marine impellers.

What does the medium need to do?

Grow the microorganism so it produces biomass and product and should not interfere with down stream processing.

Why mixing in fermentation is done?

- ✓ To disperse the air bubbles.
- ✓ To suspend the cells.

- ✓ To enhance heat and mass transfer in the medium.

Media for the fermentation? 5 Marks

Detailed investigation is needed to establish the most suitable medium for an individual fermentation process, but certain basic requirements must be met by any such medium. All micro-organisms require water, Sources of energy, Carbon, Nitrogen, Mineral elements, and possibly vitamins, plus oxygen of aerobic) such as CO2 etc.

- It will produce the maximum yield of product or biomass per gram of substrate used.
- It will produce the maximum concentration of product or biomass.
- It will permit the maximum rate of product formation
- There will be the minimum yield of undesired products
- It will be of a consistent quality and be readily available throughout the year.

Define Rheology? 2 Marks

The individual components of the medium can influence the viscosity of the final medium and its subsequent behavior with respect to aeration and agitation.

Sources of Inorganic nitrogen compounds'? 5 Marks

Inorganic nitrogen may be supplied as ammonia gas, an salts of nitrates. Ammonia has been used for PH control and as the major nitrogen source in a defined medium for the commercial production of human serum albumin by Saccharomyces cerevisiae. Ammonium salts such as ammonium sulphate will usually produce acid conditions as the ammonium ion is utilized and the free acid will be liberated. On the other hand, nitrates will normally cause an alkaline drift as they are metabolized Ammonium nitrate will first cause an acid drift as the ammonium ion is utilized, and nitrate assimilation is

repressed. When the ammonium ion has been exhausted, there is an alkaline drift as the nitrate is used as an alternative nitrogen source.

Difference between differential and selective media? 3 marks

- **Differential Media:**

Differential media make it easier to distinguish colonies of the desired organism from other colonies growing on the same plate. Similarly, pure cultures of microorganisms have identifiable reactions with differential media in tubes or plates.

- **Selective Media:**

Selective media are designed to suppress the growth of unwanted bacteria and encourage the growth of the desired microbes.

Define metabolites and what are types of metabolites?

Metabolites

Intermediates and products of metabolism. All the compounds and substances that are stops of metabolic reaction are called metabolites.

Types of Metabolites

- **Primary Metabolites**

These metabolites are those which are directly involved in the process

- **Secondary Metabolites**

These metabolites are those which are not directly involved in those metabolic processes, but usually have an important ecological function.

- **Tertiary Metabolites**

Metabolites may involve acids e.g. Citric acid, Oxalic acid, Glutamic acid and Amino acid.

Enlist the 5 commercially important fermentations? 5

There are five major groups of commercially important fermentations:

1. Those that produce microbial cells (or Biomass) as the product.
2. Those that produce microbial metabolites
3. Those that produce microbial enzymes.
4. Those that produce recombinant products
5. Those that modify a compound which is added to the fermentation – the transformation process.

How enzymes can be classified?

Enzymes can also be classified as constitutive enzymes or inducible enzymes

Constitutive enzyme:

An enzyme that is synthesized continually regardless of growth conditions.

Inducible enzyme:

An enzyme that is synthesized only in the presence of the substrate that acts as an inducer.

Define Biotransformation?

The conversion of molecules from one form to another within an organism.

Examples:

Starch → Maltose

Starch → Glucose

Benzaldehyde →LPAC

Define Medium?

The intervening substances through which metabolic requirements are conveyed or physical forces are transmitted.

Write down the two sub-processes of fermentation?

Fermentation process has two sub-processes that are:

- **Up-streaming Process (Step 1-5)**

Upstream processing is an industrial term describing the steps followed for the production of a specific product by industrial means.

- **Down-streaming Process (Step 6-7)**

Downstream processing includes the recovery of the products in a pure state and the effluent treatment. Down-streaming process product recovery is carried out through a series of operations including cell separation by settling, centrifugation or filtration. Product recovery by disruption of cells (if the product intracellular); extraction and purification of the product. Finally, the effluents are treated by chemical, physical or biological methods

Difference between intracellular or Extracellular enzymes?

Intracellular Enzymes

Enzymes are those which remain within the cell and are responsible of catalyzing all those processes need in cells.

Extracellular Enzymes

Enzymes are those which are released by the cells to perform a job external to the cell e.g. digestive enzymes.

Difference between differential media and reducing media? 3 Marks

- **Differential Media:**

Differential media make it easier to distinguish colonies of the desired organism from other colonies growing on the same plate. Similarly, pure cultures of microorganisms have identifiable reactions with differential media in tubes or plates.

Reducing media:

The cultivation of anaerobic bacteria poses a special problem because anaerobes might be killed by exposure to oxygen, special media called Reducing media must be used. Reducing media contain ingredients such as sodium Thioglycolate, that chemically combine with dissolved oxygen and deplete the oxygen in the culture medium.

Major group of important fermentation that are classified?

- Lactic acid fermentation
- Alcohol fermentation
- Acetic acid fermentation
- Butyric acid fermentation

Which ingredients are used in fermentation?

- Sugar
- Bacterial culture
- Ethanol
- Lactic acid
- Hydrogen gas (H2)
- Water
- Carbon sources and energy sources
- Nitrogen sources
- Vitamins and growth factors
- Minerals and trace elements
- Buffers
- Inducers
- Precursors
- Chelators

- Inhibitors e.g. KMS in beer medium
- Antifoams

Methods of controlling pH in fermentation?

- ✓ Keeping partial pressure of CO_2 constant
- ✓ Pumping base into the culture
- ✓ Combining pumping base and sparging gases
- ✓ Using only sparging gases